

# Data Availability and Experimental Summary

**Companion document to:** CROSS\_DOMAIN\_CIRCUIT\_PAPER.md **Project:** Cross-Domain Circuit Analysis of Factual Knowledge Retrieval in LLMs **Last updated:** 2026-04-08

This document provides a complete inventory of the data and experiments supporting the main paper's findings, intended for reproducibility and review.

## 1. Scope

- **60 attribution circuits** generated via the Neuronpedia Circuit Tracer API
- **2 models:** GEMMA-2-2B (26 layers, 16k SAE features/layer), QWEN3-4B (36 layers)
- **3 knowledge domains:** chemistry (10 prompts), geography (10 prompts), history (10 prompts)
- **80 causal steering experiments** on the GEMMA model (QWEN steering API not yet available)
- **244 cross-circuit bottleneck features** identified; 130 semantically annotated

## 2. Prompt Set

### *Chemistry (10)*

Gold, iron, lead, mercury, argon, sodium, silver, potassium, copper, tungsten — all using the format "The chemical symbol for X is".

### *Geography (10)*

France, Japan, Germany, Egypt, Italy, Spain, Canada, Thailand, Turkey, South Korea — all using the format "The capital of X is".

### *History (10)*

WWII, moon landing, Berlin Wall, Titanic, American independence, French Revolution, WWI, Columbus, Great Fire of London, Mandela release — all using the format "[Event] [occurred/ended/began] in".

Full prompt list with expected answers: Appendix A of CROSS\_DOMAIN\_CIRCUIT\_PAPER.md.

## 3. Directory Structure

All data lives under neuronpedia\_pipeline/data/ (gitignored, ~2 GB).

### *3.1 Per-Circuit Data*

```
data/prompts/<model>_<prompt-slug>/
  1_generation/           Raw API graph JSON
  2_conversion/           Standardized pipeline format
  3_analysis/             Structural analysis + traceback paths
  4_visualizations/       8 PNG figures per circuit
```

60 circuit directories total. Naming convention: gemma-2-2b\_the-chemical-symbol-for-gold-is, qwen3-4b\_the-capital-of-japan-is, etc.

### 3.2 Cross-Circuit Aggregates

|                                        |                                  |
|----------------------------------------|----------------------------------|
| data/stage_1_5_bottleneck_library.json | Cross-circuit bottleneck library |
| data/stage_1_5_cross_circuit_report.md | Summary report                   |
| data/stage_1_5_visualizations/         | Cross-circuit visualizations     |

### 3.3 Stage 2 Analyses

Each Stage 2 analysis writes to its own directory:

|                                    |                                                 |
|------------------------------------|-------------------------------------------------|
| data/stage_2_analysis/             | Main cross-domain statistical results           |
| data/stage_2_minimal_pathways/     | Essential pathway extraction (94.1% redundancy) |
| data/stage_2_multi_algorithm/      | Community detection with 4 algorithms           |
| data/stage_2_polysemanticity/      | Semantic purity analysis                        |
| data/stage_2_layer_energy/         | Per-layer activation energy profiles            |
| data/stage_2_edge_flow/            | Edge weight flow patterns                       |
| data/stage_2_coactivation/         | Feature co-activation networks                  |
| data/stage_2_output_decomposition/ | Output node contribution analysis               |
| data/stage_2_enhanced_analysis/    | Enhanced circuit metrics (8 categories)         |
| data/stage_2_statistical_deepdive/ | Extended statistical tests                      |
| data/stage_2_outlier_analysis/     | Circuit outlier identification                  |
| data/stage_2_unused_columns/       | Hidden predictor discovery                      |
| data/stage_2_annotations/          | Feature annotations                             |
| data/stage_2_figures/              | Publication-quality figures                     |
| data/stage_2_visualizations/       | Additional visualizations                       |

### 3.4 Steering Experiments

|                                          |                                                                |
|------------------------------------------|----------------------------------------------------------------|
| data/stage_3_steering/                   | D4 (20 experiments) + D5 expansion (30 experiments) = 50 total |
| data/stage_2_essential_pathway_steering/ | New 30 experiments (D6, added 2026-04-08)                      |

## 4. Experiments Summary

### 4.1 Stage 1: Initial Analysis

- Per-circuit analysis of 60 circuits with traceback graphing (backward BFS, geometric decay = 0.8)
- Bottleneck identification: features where  $\geq 60\%$  of critical paths converge
- Circuit statistics: nodes, edges, communities, centrality, layer distribution

### 4.2 Stage 1.5: Cross-Circuit Library

- Aggregated bottleneck features across all 60 circuits
- 244 unique cross-circuit features identified (all appearing in  $\geq 2$  circuits)
- 130 semantically annotated via Neuronpedia feature API (GEMMA only)

4.3 Stage 2: Cross-Domain Statistics

| Analysis                                      | Output                         | Key result                                      |
|-----------------------------------------------|--------------------------------|-------------------------------------------------|
| Cross-category regression (ANOVA, regression) | stage_2_analysis/              | 13/16 structural metrics differ cross-model     |
| Layer energy profiling                        | stage_2_layer_energy/          | Within-model cos sim=0.978; between-model=0.696 |
| Edge flow analysis                            | stage_2_edge_flow/             | 80-85% of edges are skip connections            |
| Minimal pathway extraction                    | stage_2_minimal_pathways/      | 94.1% node redundancy                           |
| Multi-algorithm community validation          | stage_2_multi_algorithm/       | Modularity >0.4, Jaccard agreement=0.363        |
| Polysemanticity classification                | stage_2_polysemanticity/       | 87.7% monosemantic; CODE+LANGUAGE=40%           |
| Co-activation analysis                        | stage_2_coactivation/          | Within-layer 1.9× cross-layer                   |
| Output decomposition                          | stage_2_output_decomposit ion/ | Output/path overlap ratio up to 4.4×            |

4.4 Stage 3: Causal Steering Validation

D4 (20 experiments) — Initial 5 features, 6 circuits

- Features: L0\_F1813559, L3\_F5150441, L4\_F110446948, L6\_F2586668, L24\_F88478228
- Strength: ±20
- 50% text change rate; mean logprob shift = 0.045

D5 (30 experiments) — Frequency-based expansion (5 new features)

- Features: L1\_F99962728, L2\_F25751073, L5\_F7993995, L7\_F4828270, L9\_F125286525
- Strength: ±20
- 23.3% text change rate; highest-frequency feature produced zero changes

D6 (30 experiments) — Essential-pathway features (added 2026-04-08)

- Features: L0\_F64712375, L1\_F1736314, L21\_F5479683, L25\_F50014975, L24\_F18002975
- Strength: ±20
- 26.7% text change rate; mean KL divergence = 1.448 (highest of any group)
- Domain breakdown: chemistry 0%, geography 20%, history 60%

Aggregate (80 experiments): three-tier dissociation finding — topology predicts distributional perturbation, frequency predicts statistical commonality, output determinism governs text-level changes.

5. Key Result Files

Files referenced by specific findings in the paper:

| Finding                        | File                                                                       |
|--------------------------------|----------------------------------------------------------------------------|
| Bottleneck library             | stage_1_5_bottleneck_library.json                                          |
| Minimal pathway metrics        | stage_2_minimal_pathways/minimal_pathway_results.json                      |
| Multi-algorithm validation     | stage_2_multi_algorithm/multi_algorithm_results.json                       |
| Layer energy profiles          | stage_2_layer_energy/layer_energy_analysis.json                            |
| Steering D4 results            | stage_3_steering/steering_results.json                                     |
| Steering D4+D5 results         | stage_3_steering/steering_results.json                                     |
| Steering D6 results            | stage_2_essential_pathway_steering/essential_pathway_steering_results.json |
| Polysemanticity classification | stage_2_polysemanticity/polysemanticity_results.json                       |

## 6. Reproducibility

### Requirements

- Python 3.8+, dependencies in neuronpedia\_pipeline/config/requirements.txt
- Neuronpedia API key (free at neuronpedia.org)
- ~2 GB disk for full pipeline outputs

### Reproducing a single circuit

```
python run_full_pipeline.py --prompt "The capital of France is" --model gemma-2-2b
```

### Reproducing a steering experiment set

```
python scripts/5_steering_validation.py --quick           # D4 equivalent
python scripts/stage_2_expanded_steering.py              # D5 equivalent
python scripts/stage_2_essential_pathway_steering.py     # D6 equivalent
```

### Regenerating Stage 2 statistical analyses

```
python scripts/stage_2_cross_category_analysis.py
python scripts/stage_2_layer_energy_analysis.py
python scripts/stage_2_minimal_pathways.py
python scripts/stage_2_multi_algorithm_validation.py
python scripts/stage_2_polysemanticity_analysis.py
```

Each script reads from data/prompts/ and writes to its own data/stage\_2\_\*/ directory.

## 7. Limitations of Released Data

1. **Generated data is gitignored** due to size (~2 GB). Recreate by running the pipeline against the prompt list in Appendix A.
2. **Neuronpedia API key** required for generation and semantic annotation — free but rate-limited.
3. **QWEN feature API** not yet publicly available; QWEN circuits are structural only.

4. **Manual semantic annotations** (80 features) are in `semantic_taxonomy_annotations_auto.csv` in the repo root.

5. **Steering experiments** are non-deterministic at the token level but use `temperature=0`, `seed=42` for reproducibility.

## 8. Contact and Citation

Paper: `CROSS_DOMAIN_CIRCUIT_PAPER.md` Repository: `github.com/J-Lawrence10/autocircuit`  
(branch: `main`) Related work: Anthropic's attribution graph framework; Neuronpedia platform (Decode Research)